1.

1.1. Used Polars for this Question

Most airports

code	name	airport_count
str	str	u32
"US"	"United States"	25170
"BR"	"Brazil"	5546
"CA"	"Canada"	2829

Least airports (I printed all that had a rank <= 10 but most were equal at 1)

```
pl.Config.set_tbl_rows(-1)
bottom_filter =  pl.col("airport_count").rank(method="min",descending=False) <= 10
countries_by_airports.filter(bottom_filter).sort('airport_count', descending = True)
```

code	name	airport_count
str	str	u32
"BL"	"Saint Barthéle...	1
"CC"	"Cocos (Keeling...	1
"CW"	"Curaçao"	1
"CX"	"Christmas Isla...	1
"GI"	"Gibraltar"	1
"GM"	"Gambia"	1
"IO"	"British Indian...	1
"JE"	"Jersey"	1
"NF"	"Norfolk Island...	1
"NR"	"Nauru"	1
"NU"	"Niue"	1
"SX"	"Sint Maarten"	1
"VA"	"Vatican City"	1
"YT"	"Mayotte"	1
"GS"	"South Georgia ...	0
"PN"	"Pitcairn"	0
"TK"	"Tokelau"	0

1.2.

id	name	iso_country	length_ft	width_ft
i64	str	str	i64	i64
46491	"Pontiac Airpar...	"CA"	120000	1000
22159	"Gunflint Seapl...	"US"	30000	5000
30788	"Qamdo Bangda A...	"CN"	18045	148
26407	"Ulyanovsk East...	"RU"	16404	323
2847	"Pierre Van Ryn...	"ZA"	16076	197

2. Used bash (curl command)

```
azure_url = (
    f'''https://{upload_info['account_name']}.blob.core.windows.net'''
    f'''/{upload_info['blob_container']}/{upload_info['blob_path']}'''
    f'''/{upload_info['file']}'''
    f'''?{upload_info['share_signature']}'''
)
azure_url
```

```
'https://sacodeassessment.blob.core.windows.net/results//ingest-assessment-20260301-ms/longest_runways_per_country.csv?sv=2022-11-02&ss=
b&srt=o&sp=wc&se=2034-11-11T11:00:00Z&st=2024-11-10T23:00:00Z&spr=https&sig=D%2BgRbWPJDTmsbPtyfTEiTnb7gg594uNsvm62oQK49Yg%3D'
```

```
!curl -X PUT "{azure_url}" --data-binary @"{upload_info['file']}" -H "x-ms-blob-type: BlockBlob" -v
```

```
* Host sacodeassessment.blob.core.windows.net:443 was resolved.
* IPv6: (none)
* IPv4: 20.209.73.130
*   Trying 20.209.73.130:443...
```

3. Couldn’t get a reply for any country with timeout(1,5). Also couldn’t post the revenue.
   3.1. so , all countries are missing

AD: Unexpected ConnectTimeout: HTTPSConnectionPool(host='code001.ecsbdp.com', port=443): Max retries exceeded with url: /get-country-info/countries/AD?client=abc123 (Caused by ConnectTimeoutError(<urllib3.connection.HTTPSConnection object at 0x166397910>, 'Connection to code001.ecsbdp.com timed out. (connect timeout=1)'))

4.

4.1. Used polars with psycopg2-binary (to execute the queries)

is_jazz_customer	total_sales
i64	decimal[*,2]
0	9762.74
1	11085.88

First finding the relationship and then query

```
desc = pl.read_database("""
  SELECT table_name, column_name, data_type, is_nullable
  FROM information_schema.columns
  WHERE table_schema = 'public'
  ORDER BY table_name, ordinal_position
""", conn)
desc.head()
```

shape: (5, 4)

table_name	column_name	data_type	is_nullable
str	str	str	str
"album"	"album_id"	"integer"	"NO"
"album"	"title"	"character vary...	"NO"
"album"	"artist_id"	"integer"	"NO"
"artist"	"artist_id"	"integer"	"NO"
"artist"	"name"	"character vary...	"YES"

```
pl.read_database(
    """
    WITH track_type AS (
        SELECT
            track_id
            , CASE WHEN g.name = 'Jazz' THEN 1 ELSE 0 END AS is_jazz
        FROM track AS t
        LEFT JOIN genre AS g USING (genre_id)
    )
    , customer_sales_by_type AS (
        SELECT
            customer_id
            , MAX(is_jazz) AS is_jazz_customer
            , SUM(total) total_sales
        FROM invoice AS i
        LEFT JOIN invoice_line AS il USING(invoice_id)
        LEFT JOIN track_type AS tt USING(track_id)
        GROUP BY customer_id
    )
    SELECT
        is_jazz_customer
        , SUM(total_sales) AS total_sales
    FROM customer_sales_by_type
    GROUP BY is_jazz_customer
    """
    , conn
)
```

4.2.

4.2.1. Materialization of Join with downside of taking care of adding new rows when underlying tables changes, as well storage cost.

4.2.2. Indexing which is more logical since underlying tables change infrequently (no refresh overhead). Could use something like:
CREATE INDEX idx_track_album_lower ON track (album_id, LOWER(name) varchar_pattern_ops)

5. Got to search this one
It's WT (JSON Web Token)
A JWT has three Base64URL-encoded parts: header, payload (claims), and signature
In Payload we have iss (issuer), aud (audience), exp (expiry), appid, user id, etc. and
exp: a timestamp in March 2021 → long expired relative to 2026

```
=== HEADER ===
{'typ': 'JWT', 'alg': 'RS256', 'x5t': 'nOo3ZDrODXEK1jKWhXslHR_KXEg', 'kid': 'nOo3ZDrODXEK1jKWhXslHR_KXEg'}

=== PAYLOAD ===
{'aud': 'https://Eneco.onmicrosoft.com/ecsaz-apigee-odp-t', 'iss': 'https://sts.windows.net/eca36054-49a9-4731-a42f-8400670fc02
2/', 'iat': 1616416705, 'nbf': 1616416705, 'exp': 1616420605, 'aio': 'E2ZgYEg3Vdmf6hspv/uPMbff8/MCAA==', 'appid': '1e0fb354-a78
d-4f5b-9966-ed623a624599', 'idp': 'https://sts.windows.net/eca36054-49a9-4731-a42f-8400670fc022/', 'oid': 'db6e89f9-db16-4252-9
429-c4d5e48adabd', 'sub': 'db6e89f9-db16-4252-9429-c4d5e48adabd', 'tid': 'eca36054-49a9-4731-a42f-8400670fc022', 'uti': 'SgBa7Q
3KTUy3Qf8hzOY3AA', 'ver': '1.0'}
Expiry: 2021-03-22 14:43 UTC
Issuer: https://sts.windows.net/eca36054-49a9-4731-a42f-8400670fc022/
Audience: https://Eneco.onmicrosoft.com/ecsaz-apigee-odp-t
App ID: 1e0fb354-a78d-4f5b-9966-ed623a624599

=== FULL DECODE ===
{'aud': 'https://Eneco.onmicrosoft.com/ecsaz-apigee-odp-t', 'iss': 'https://sts.windows.net/eca36054-49a9-4731-a42f-8400670fc02
2/', 'iat': 1616416705, 'nbf': 1616416705, 'exp': 1616420605, 'aio': 'E2ZgYEg3Vdmf6hspv/uPMbff8/MCAA==', 'appid': '1e0fb354-a78
d-4f5b-9966-ed623a624599', 'idp': 'https://sts.windows.net/eca36054-49a9-4731-a42f-8400670fc022/', 'oid': 'db6e89f9-db16-4252-9
429-c4d5e48adabd', 'sub': 'db6e89f9-db16-4252-9429-c4d5e48adabd', 'tid': 'eca36054-49a9-4731-a42f-8400670fc022', 'uti': 'SgBa7Q
3KTUy3Qf8hzOY3AA', 'ver': '1.0'}
```

6.

6.1. One on top of the page

```
!echo "RkxBR3s3Mjc0ZTVkYy1hYjYwLTRmYzItOGY1MS1lZTZlY2ViNmJlZjd9" | base64 -d
```

```
FLAG{7274e5dc-ab60-4fc2-8f51-ee6eceb6bef7}
```

6.2. One in the countries.csv

```
%%bash -s "$urls_str"
read -a urls <<< "$1"
for url in "${urls[@]}"; do
  fn=$(basename "$url")
  curl -O "$url" && echo "✅ $fn"
  echo "=== $fn flags ==="
  grep -i 'FLAG{' "$fn"
  grep 'RkxBR3s' "$fn"
  echo "---"
done
```

```
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 9127k  100 9127k    0     0  8226k      0  0:00:01  0:00:01 --:--:-- 8229k
✅ airports.csv
=== airports.csv flags ===
---
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 20707  100 20707    0     0   240k      0 --:--:-- --:--:-- --:--:--  240k
✅ countries.csv
=== countries.csv flags ===
302629,"CY","Cyprus","AS","https://en.wikipedia.org/wiki/Cyprus","FLAG{e9cade1a-55a8-4af5-887d-a70ac0f3078a}"
---
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 2984k  100 2984k    0     0  7744k      0 --:--:-- --:--:-- --:--:-- 7732k
✅ runways.csv
=== runways.csv flags ===
---
```